From Buildout Story to Backlash Story: Media Sentiment Toward Data Centers in a Six-Technology Benchmark, 2015–2026

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Abstract

English-language news coverage of data centers grew tenfold between January 2022 and June 2026 while its sentiment reversed sign. Using two complementary GDELT sampling frames (335,206 fulltext-mention articles and a census of 103,341 articles with the term in the headline frame), LLM classification of 8,100 sampled headlines, and a six-technology comparison corpus of 921,170 articles spanning 2015–2026, we measure the reversal and place it in context. Net headline sentiment toward data centers fell from +54 points in 2022 to −5 points in the first half of 2026, with the first net-negative month in March 2026. Among six infrastructure and technology buildouts (5G, wind, solar, fracking, cryptocurrency mining, data centers), the data center decline is the largest sustained tone decline from a net-positive starting level, and unlike the 5G and wind episodes, which recovered within two to four months of turning negative, it had not recovered as of June 2026. Classifying outlets by type shows the negativity is stratified: local news outlets run roughly 13 percentage points more negative than national outlets throughout, and are the only outlet class whose net sentiment turned sustainably negative; local and national series move together with no detectable lead-lag. Thematically, coverage shifted from investment and buildout (59% of headlines in 2022, 27% in 2026) toward community opposition (5% to 22%) and energy, while water-related coverage grew 2.4x from a small base. All data, code, prompts, and classification rubrics are released for reproduction.

1 Introduction

Between 2022 and 2026, data centers moved from a specialist infrastructure topic to a mainstream news subject. The share of global English-language news coverage mentioning data centers grew 7.2x over the period, and the absolute number of articles carrying the term in their headline frame grew from 838 to 9,120 per month. Over the same window, the sentiment of that coverage reversed: headlines that in 2022 were dominated by investment announcements were, by mid-2026, dominated by stories about community opposition, grid strain, and environmental resources.

This paper measures that reversal quantitatively and asks three questions. First, how large and how fast is the sentiment decline, measured consistently over four and a half years? Second, how does it compare with prior technology buildouts that attracted public controversy: 5G, wind power, solar power, hydraulic fracturing, and cryptocurrency mining? Third, what is the internal structure of the decline across outlet types and coverage themes?

Our contributions are methodological and empirical. Methodologically, we combine two GDELT sampling frames with LLM headline classification, publish the full rubric and quality-control results, calibrate the LLM series against GDELT's independent lexicon-based tone measure, and apply a uniform promotional-content filter across all six technologies. Empirically, we document (i) a +54 to −5 point net-sentiment reversal for data centers, (ii) its position as the largest net-positive-to-negative tone decline among the six technologies studied, (iii) a 13-point stratification between local and national outlet negativity with no detectable lead-lag between them, and (iv) a thematic rotation from buildout coverage toward community opposition, energy, and, from a small base, water.

We deliberately frame the analysis descriptively. News sentiment is an observable property of published coverage; we make no claims about the accuracy of that coverage or the merits of the underlying projects, and the patterns documented here are offered as measurements that any stakeholder, including infrastructure developers, utilities, regulators, and journalists, can use.

2 Data and methods

2.1 Two GDELT frames

All raw data come from the GDELT Project [1]. We use two frames with different tradeoffs between recall and precision:

- **Fulltext frame.** All English-language articles matching "data center(s)" or "datacenter(s)" anywhere in the article, via the GDELT DOC 2.0 API: 335,206 articles, January 2022 through June 2026. This frame has high recall but includes articles that mention data centers only in passing.
- **Headline frame.** A census of articles whose URL slug contains the term, built from the GDELT Global Knowledge Graph on BigQuery (English-only; pattern `data[-_]?cent(er|re)` applied to the URL path): 103,341 unique articles over the same period. Requiring the term in the headline frame selects articles that are *about* data centers.

The two frames agree well on volume dynamics (monthly volume correlation $r = 0.93$), and the headline frame's share of the fulltext frame rose from 16% in 2022 to 48% in the first half of 2026: when data centers appear in coverage at all, they are increasingly the story rather than the backdrop.

2.2 Headline sampling and LLM classification

From the headline census we drew a seeded random sample of 150 articles per month (8,100 total across 54 months; median 141 relevant per month after filtering). Headline text was recovered by fetching live page titles (72% of the sample) or reconstructing them from URL slugs (28%; 80% in 2026 as older pages disappear, a limitation discussed in Section 5).

Each headline was classified by Claude Sonnet (Anthropic) [3] against a committed rubric with three dimensions: relevance (94% of sampled headlines are relevant), sentiment toward data centers (positive, negative, neutral), and one of eight themes (energy and grid; environment and water; community opposition; investment and buildout; jobs and economy; policy and regulation; technology and operations; other). The full rubric, prompts, and per-headline labels are in the repository.

Quality control. A seeded 10% subsample (400 headlines) was re-classified independently with shuffled batch composition. Agreement was 98.8% on relevance, 87.2% on sentiment, and 92.8% on theme. No disagreement flipped a headline between positive and negative (all sentiment disagreements involved the neutral class), and the resulting monthly net-sentiment series differ by 0.8 percentage points on average.

Table 1: Correlation between LLM net headline sentiment and article-count-weighted GDELT tone. Quarterly aggregation; the data center series is also shown monthly.

Technology	Periods	r
Data centers (monthly)	54	0.85
Data centers (quarterly)	18	0.92
Cryptocurrency mining	46	0.66
5G	46	0.60
Wind power	46	0.55
Solar power	46	0.38
Hydraulic fracturing	46	−0.04

2.3 Six-technology comparison corpus

To place the data center episode in context we built an analogous headline-frame corpus for five other technologies over 2015–2026: 5G (276,950 articles), wind power (227,118), solar power (108,798), hydraulic fracturing (87,229), and cryptocurrency mining (26,033), for a comparison corpus of 921,170 articles including data centers. For each technology we classified a quarterly headline sample with the same rubric and pipeline (1,827–1,840 headlines per technology; 9,187 total), and computed the GDELT lexicon tone of every article in the corpus.

Promotional-content filter. Press-release wires and promotional content are unevenly distributed across technologies and time; unfiltered, they produced a spurious positive tone spike in 2025 cryptocurrency coverage of which 22% proved promotional. We therefore excluded PR-wire and promotional content uniformly across all six technologies (the cryptocurrency example is the largest correction: article-count-weighted mean tone +1.03 raw versus +0.09 filtered). Raw series are retained in the repository as an annex. The data center finding is strengthened, not weakened, by the filter.

2.4 Calibration against GDELT tone

The LLM net-sentiment series and GDELT's lexicon-based tone are independent measurements: the former reads 150 sampled headlines per month, the latter averages a dictionary score over every article in the census. Their agreement is a check on both. Table 1 reports Pearson correlations computed by the committed calibration script.

Calibration is strongest where sentiment actually moves: the data center series, with the largest dynamic range, correlates at $r = 0.92$ quarterly. Fracking, whose tone is persistently negative with little variation over 2015–2026, shows no correlation; when a series spends a decade in a flat regime, sampling noise dominates both measures and correlation is not an informative validity check. We therefore read Table 1 as validating the method in exactly the regimes where we use it to make claims about change.

2.5 Outlet-type classification

For the outlet-structure analysis (Section 3.3) we classified all 1,979 unique domains in the classified sample into five outlet types (local, national, trade, wire/PR, other) using an LLM pass over the domain name and sample headlines, against a committed rubric. Locality-sounding PR-syndication networks were identified via duplicate press-release headlines across domains and assigned to wire/PR. An independent 10% double-classification (197 domains) agreed 92.9% overall

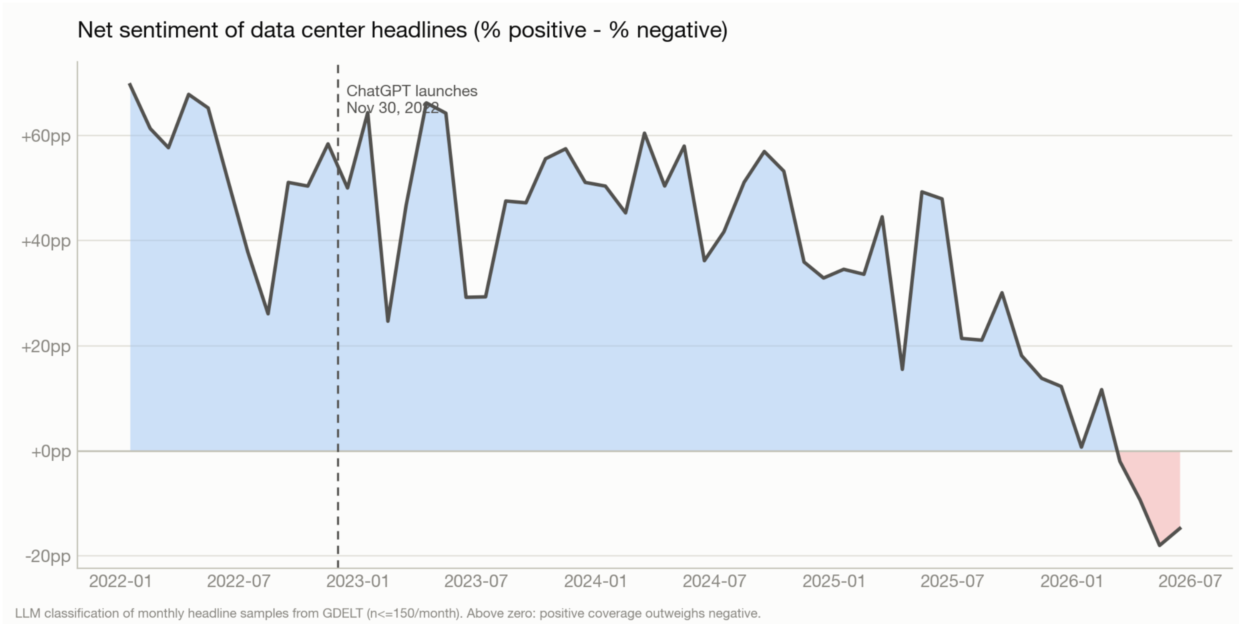


Figure 1: Net sentiment of data center headlines (share positive minus share negative), monthly, from LLM classification of 150 sampled headlines per month. The series first turns negative in March 2026.

and 100% on the local-versus-not distinction that the analysis depends on. Wire/PR (1,716 relevant headlines) and unclassifiable (93) rows are excluded from outlet-level series, leaving 1,980 local, 1,434 national, and 2,387 trade relevant headlines.

3 Results

3.1 The reversal

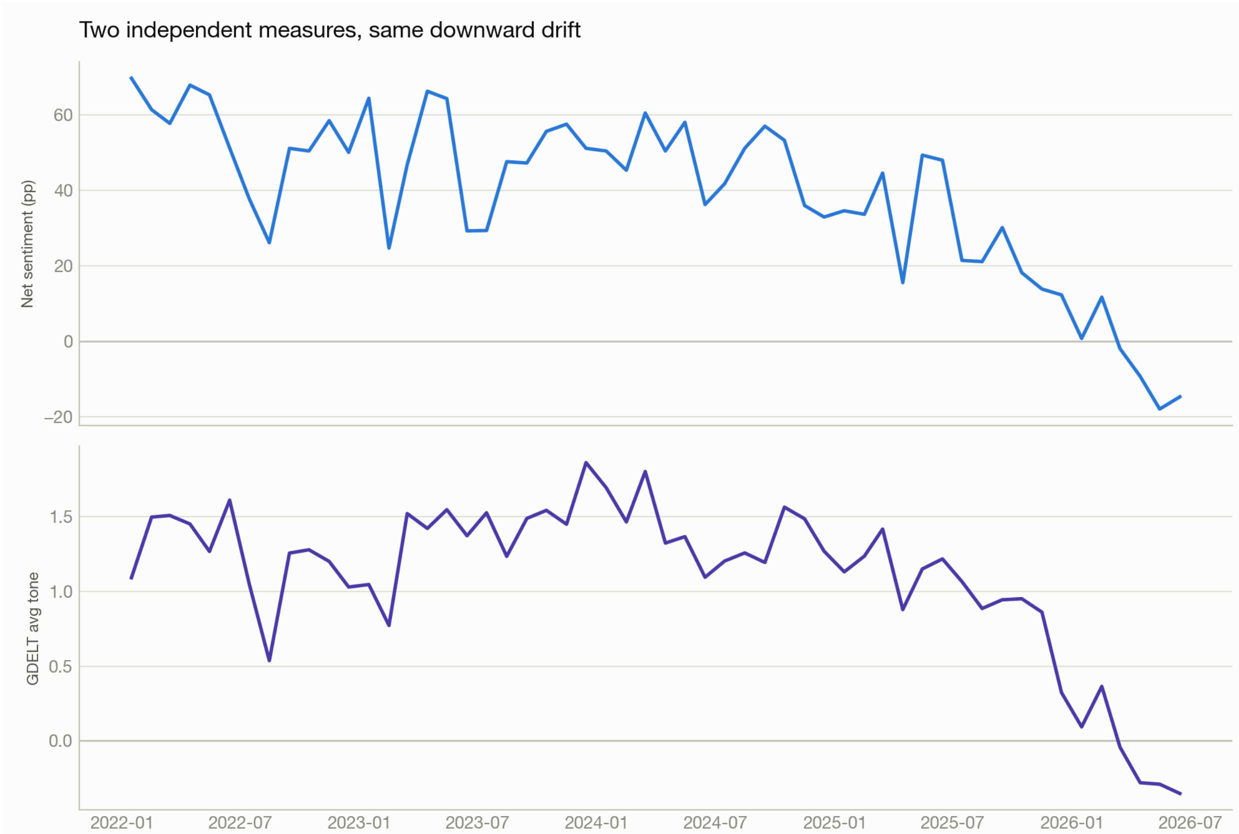
Net headline sentiment (share positive minus share negative, among relevant headlines) declined monotonically at annual resolution: +54 points in 2022, +49 in 2023, +48 in 2024, +28 in 2025, and −5 in the first half of 2026 (Figure 1). The first net-negative month was March 2026; the deepest month in the series is May 2026 at −18 points. The series peaked in January 2022 at +70 points.

GDELT's independent lexicon tone over the full census confirms the shape: tone fell from +1.09 in January 2022 to −0.35 in June 2026, first crossing zero in March 2026, the same month as the LLM series (Figure 2). Coverage volume grew 10.9x over the same window (838 to 9,120 headline-frame articles per month), so the negative turn is not a thinning of coverage but a change in the composition of a rapidly expanding coverage base.

3.2 Six-technology benchmark

Figure 3 aligns the six technologies' tone series around their coverage takeoffs, and Figure 4 summarizes the timing. We define *takeoff* as the first month in which a three-month coverage window sustains at least twice the median of all prior months (minimum 500 articles), and a *flip* as two consecutive smoothed months of negative tone following the pre-decline goodwill peak.

Three patterns emerge. First, the magnitude: measured as the largest 18-month decline in smoothed tone, data centers fell 1.42 tone points from a +1.26 starting level. Cryptocurrency



Top: LLM-classified headline sentiment (150/month sample). Bottom: GDELT lexicon tone across all 103,341 datacenter-headlined articles (population-wide, independent of the LLM).

Figure 2: Cross-validation: LLM net sentiment (top, monthly sample) and GDELT lexicon tone over all 103,341 census articles (bottom). The measures are independent; both cross zero in March 2026.

mining's raw decline is nominally larger (1.73 points) but starts from an approximately neutral level (-0.03); the data center episode is the largest decline from a clearly net-positive starting level, i.e., the deepest measured loss of prior goodwill among the six.

Second, the timing: data center tone flipped negative 26 months after its goodwill peak. The comparable figures are 28 months for 5G and 38 months for wind; fracking was net-negative throughout the observation window and solar never flipped.

Third, and most distinctive, the recovery: 5G tone recovered to positive four months after flipping, and wind two months after. Data center tone had not recovered as of June 2026, the end of our window. Among the six technologies, only fracking (persistently negative) offers a precedent for a flip that is not reversed within a few months (cryptocurrency mining recovered after five), and neither fracking nor cryptocurrency entered its decline from a strongly positive baseline. Whether the data center series follows the 5G/wind pattern (fast reversion) or establishes a new persistent regime is, at this writing, an open empirical question that the released pipeline can track.

3.3 Outlet structure: a level gap, not a diffusion lag

A natural hypothesis is that negativity starts in local outlets covering siting disputes and later diffuses to national coverage. The data support half of that picture. Local outlets are substantially more negative throughout: their mean quarterly negative share is 30.8% versus 18.0% for national outlets, and in 2026-Q2 the gap was 52.2% versus 31.5% (trade press: 27.8%). Local outlets are

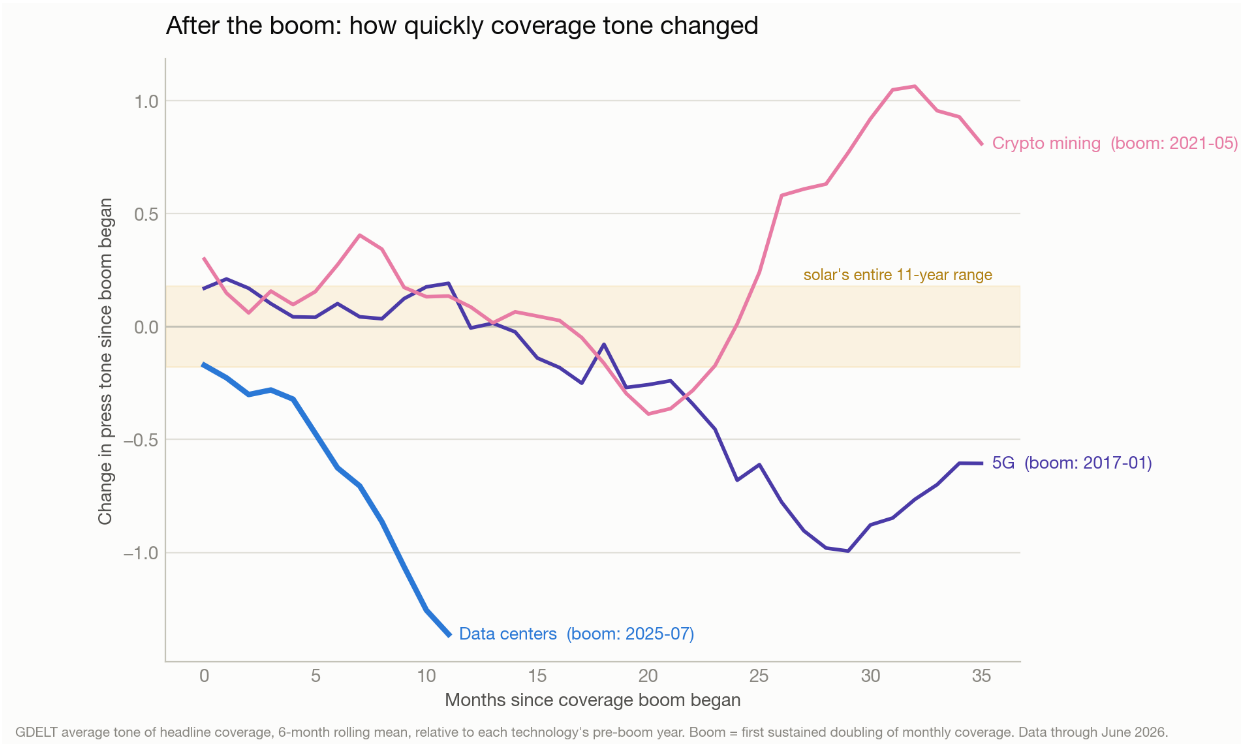


Figure 3: GDELT tone aligned on coverage takeoff for the three technologies with a detected takeoff (data centers, 5G, cryptocurrency mining), with solar's full-period range shown as a band for scale; promotional content removed uniformly. Wind, solar, and fracking never met the takeoff threshold.

also the only class whose net sentiment flipped sustainably negative (2025-Q3); national and trade net sentiment remained positive through 2026-Q2 (Figure 5).

However, we find no evidence of a lead-lag: the cross-correlation between local and national negative shares peaks at lag zero ($r = 0.74$ quarterly and $r = 0.68$ monthly, in each case exceeding every realization of an exhaustive circular-shift null, i.e. $p < 0.06$ quarterly and $p < 0.02$ monthly at the test's resolution), with symmetric decay at adjacent lags. At monthly resolution, local and national negativity move together; what distinguishes local coverage is its level, not its timing. For anyone monitoring coverage as a leading indicator, this suggests watching how negative local coverage *is*, rather than expecting it to provide months of advance warning of national coverage.

3.4 Thematic rotation, and water

The theme mix explains what changed. Investment-and-buildout headlines fell from 59% of relevant coverage in 2022 to 27% in 2026. Community opposition rose from 5% to 22%, and energy-and-grid coverage peaked at 29% of headlines in October 2024. Among negative headlines specifically, community opposition is the largest theme in both the early (2022–2023) and late (2025 through June 2026) periods, followed in the late period by energy and grid.

Water-related coverage warrants separate treatment because it is measured here two ways (Figure 6). As a classified theme (single label per headline), environment-and-water holds a roughly flat 5–6% share across the period; but because each headline receives exactly one theme, the rapid growth of the energy theme mechanically crowds the water label in stories that touch both. Keyword measurement over the full census avoids this: the share of census articles with water-related

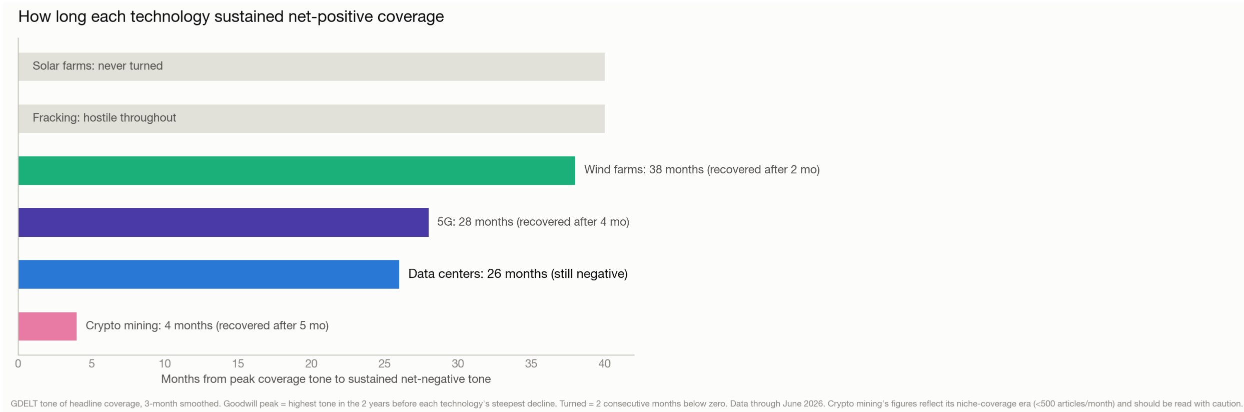


Figure 4: Months from pre-decline goodwill peak to sustained negative tone (flip), and from flip to recovery where applicable.

terms in the headline slug (water, drought, aquifer, groundwater, wastewater) grew from 1.4% in 2022–2023 to 3.2% in 2025 through June 2026, a 2.4x increase, slightly faster than the parallel energy-keyword share (2.1x) though from a base roughly five times smaller. Water is the third-ranked theme among negative headlines in both periods, behind community opposition and energy. The keyword proxy was validated against fetched titles (recall 0.95, precision 0.83, with most nominal false positives traced to title-fetch failures rather than slug errors).

4 Robustness

Promotional-content filter. As described above, the uniform filter corrects a 22%-promotional spike in 2025 cryptocurrency coverage and strengthens the data center finding.

Frame drift. Late-period cryptocurrency coverage drifts toward AI and data center usage of former mining facilities, which blurs that comparison series after 2024; we caveat rather than correct this, as any correction would require content-level judgments outside the rubric.

Independent preliminary sample. A preliminary fulltext-frame sample (January–October 2022, about 30 relevant headlines per month) yields a 2022 net sentiment of +51 points versus the census frame's +54, indicating the headline result is not an artifact of frame choice.

Measurement independence. The LLM and lexicon measures agree at $r = 0.92$ quarterly on the series where change occurs (Section 2.4).

5 Limitations

Headlines are not articles: the classifier sees the headline and domain only. The headline frame requires the term in the URL slug, excluding stories about AI infrastructure that never name data centers. Coverage is English-only. GDELT's source list over-represents wire and press-release distributors, which we mitigate but cannot eliminate with the promotional filter. For 28% of sampled headlines (80% in 2026) the title was reconstructed from the URL slug; quality control found no systematic sentiment effect, but confidence in 2026 monthly values is accordingly lower. GDELT itself had outages on 2023-03-23 and 2025-06-15 through 2025-07-01. Finally, LLM classification is imperfect (87.2% sentiment agreement on re-classification); we publish all labels, prompts, and rubrics so that any step can be audited or re-run.

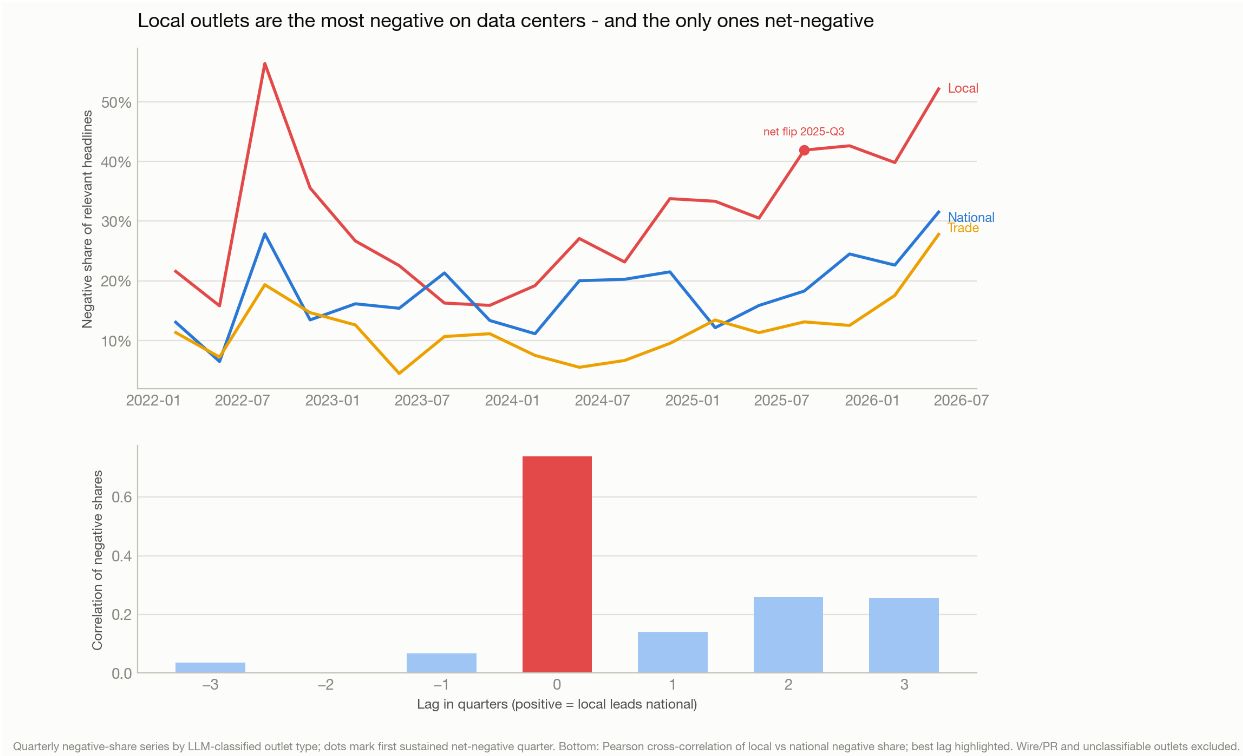


Figure 5: Top: quarterly negative share of relevant headlines by outlet type; the dot marks the first sustained net-negative quarter (local only). Bottom: cross-correlation of local versus national negative share by lag; the maximum is at lag zero.

6 Discussion

Read together, the results describe a technology buildout whose coverage grew an order of magnitude while rotating from an investment story to a community-impact story. The six-technology benchmark provides the relevant context: turning negative is not unusual (four of six technologies did), but the combination of depth (largest decline from a net-positive baseline), breadth (the only technology whose local-outlet coverage went sustainably net-negative in our outlet analysis), and duration (four negative months and counting at the series end, where 5G and wind reverted within two to four) is so far unique to data centers in this sample.

For infrastructure developers and their counterparties, the measurements suggest three monitorable quantities: the recovery clock (whether national sentiment reverts on the 5G/wind timescale or persists), the level of local negative share (which reached 52% in 2026-Q2 and moves contemporaneously with national coverage), and water salience (small but growing faster than any other environmental keyword we track). We offer these as descriptive benchmarks rather than forecasts; the released pipeline updates monthly as GDELT publishes.

Note added (July 19, 2026)

Three developments after the close of our data window (June 2026) bear directly on the measured trajectory and are noted here for completeness, with sources. On July 18, 2026, 142 demonstrations concerning data center construction were held across 42 U.S. states in a coordinated national day of protest organized by the group HumansFirst, with participants prominently citing water and

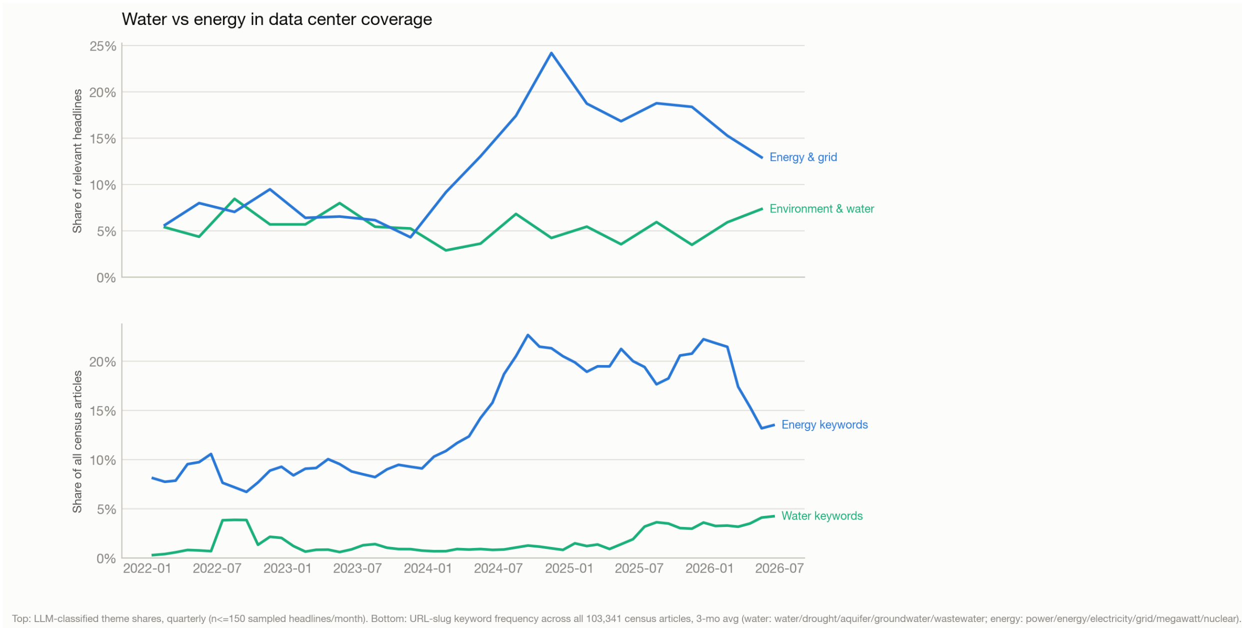


Figure 6: Water versus energy in data center coverage. Top: classified theme shares (quarterly). Bottom: keyword frequency in headline slugs across the full census (monthly, smoothed).

electricity use [4]. A Reuters/Ipsos poll fielded in June 2026 ($n = 4{,}531$) found 14% of American respondents comfortable with a data center being built near them, and 57% saying they would oppose one in their community [5]. And on July 14, 2026, New York announced the first statewide moratorium on new hyperscale (50 MW and larger) data centers, for up to one year pending a programmatic environmental review [6]. These events are consistent with the trajectory measured here: community opposition as the largest negative theme, local-outlet negative share above 50%, and rising water salience. They fall outside our measurement window and are not reflected in any series in this paper.

Data and code availability

All code, SQL, prompts, classification rubrics, per-headline labels, and processed series are available at `https://github.com/ketos7/datacenter-media-sentiment`. Code is MIT licensed; datasets are derived from the GDELT Project and released under CC BY 4.0 with attribution to GDELT [2]. All sampling and quality-control randomness is seeded, and seeds are committed.

References

[1] K. Leetaru and P. A. Schrodt, "GDELT: Global data on events, location, and tone, 1979–2012," ISA Annual Convention, San Francisco, April 2013.

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[3] Anthropic, "Claude Sonnet" (large language model), `https://www.anthropic.com/` (2026).

[4] Reuters, "Data center opponents stage 142 protests across 42 US states," July 18, 2026.

[5] Ipsos, "AI data centers are unpopular with most Americans," Reuters/Ipsos poll, June 2026, https://www.ipsos.com/en-us/ai-data-centers-are-unpopular-most-americans.

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